CLUSTERING

CLUSTERING ALGORITHMS – PART I

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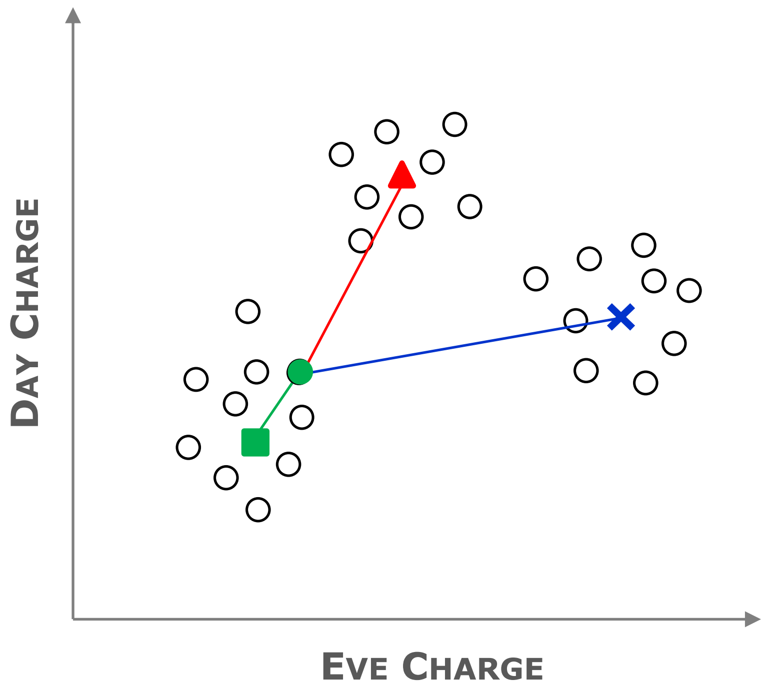
Clustering Algorithms

The following concepts will be introduced:

- ✓ **Rationale of prototype-based clustering**
- ✓ **Centroid**
- ✓ **K-Means clustering**
  - Algorithm

Prototype-Based Clustering

In **Prototype-Based clustering**, a cluster is a subset of objects (records) such that any object is closer to the prototype that defines the cluster to which it belongs to than to the prototype of any other cluster.



Different types of prototype-based clustering algorithms exist depending on the following characteristics:

- ✓ Objects belong to a single cluster
- ✓ Objects are allowed to belong to more than one cluster
- ✓ A cluster is modeled as a probability distribution
- ✓ Clusters are constrained to have fixed relationships (neighborhood relationships)

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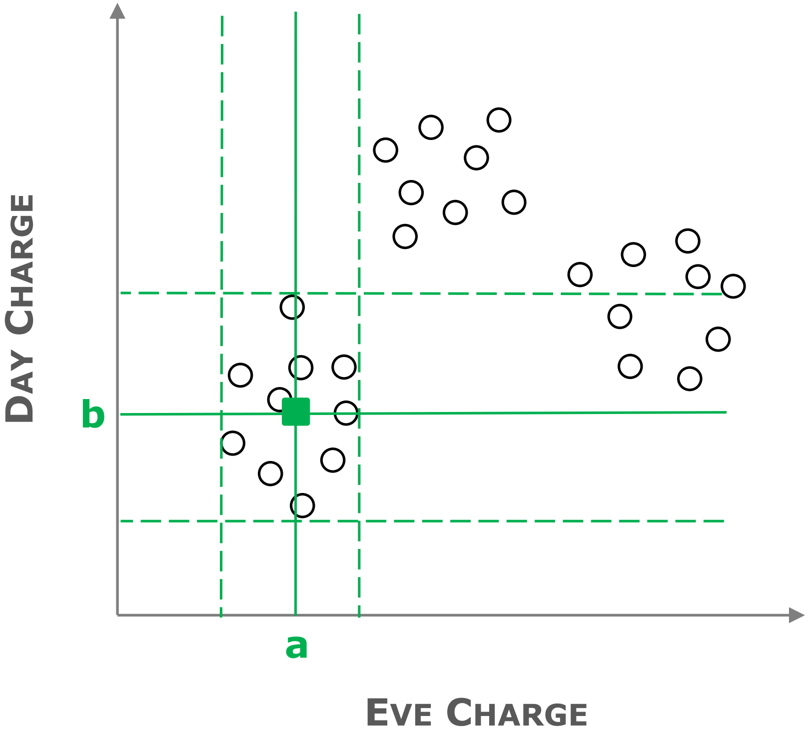

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PROTOTYPE-BASED CLUSTERING: K-MEANS

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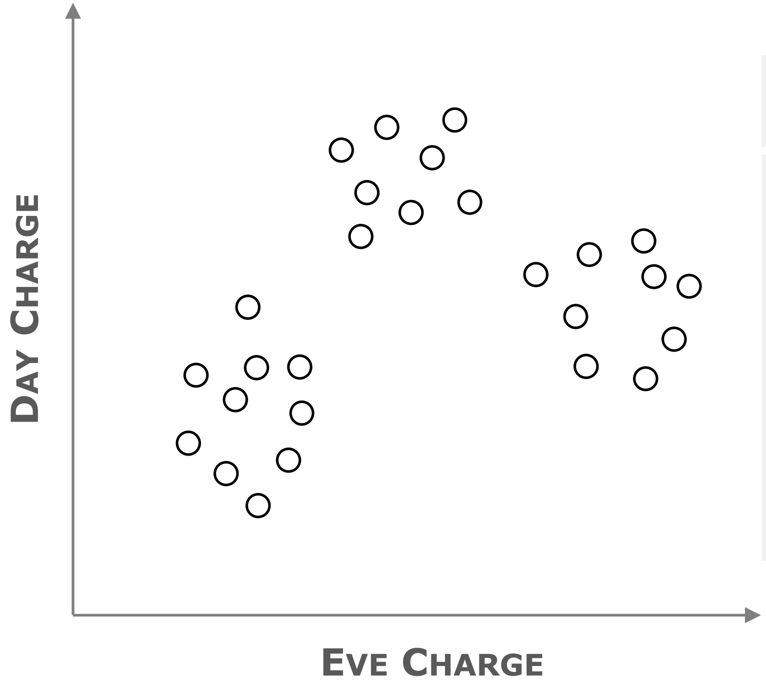

K-MEANS ALGORITHM

1. Select *K* objects (records) as centroids
2. **REPEAT**
3. Form *K* clusters, assign each record to the closest centroid
4. Recompute the centroid of each cluster
5. **UNTIL** centroids do not change

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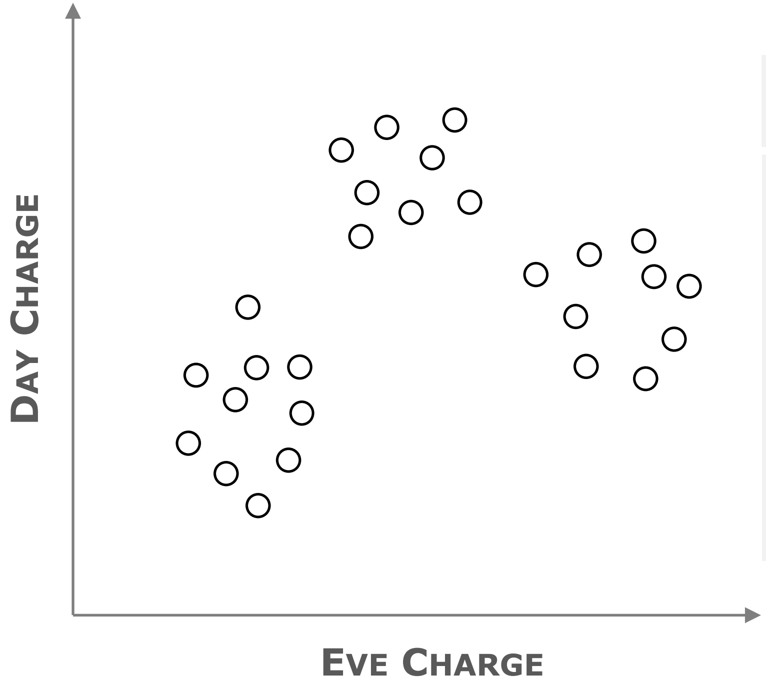
Let K be equal to 3



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1. Select K objects (records) as centroids
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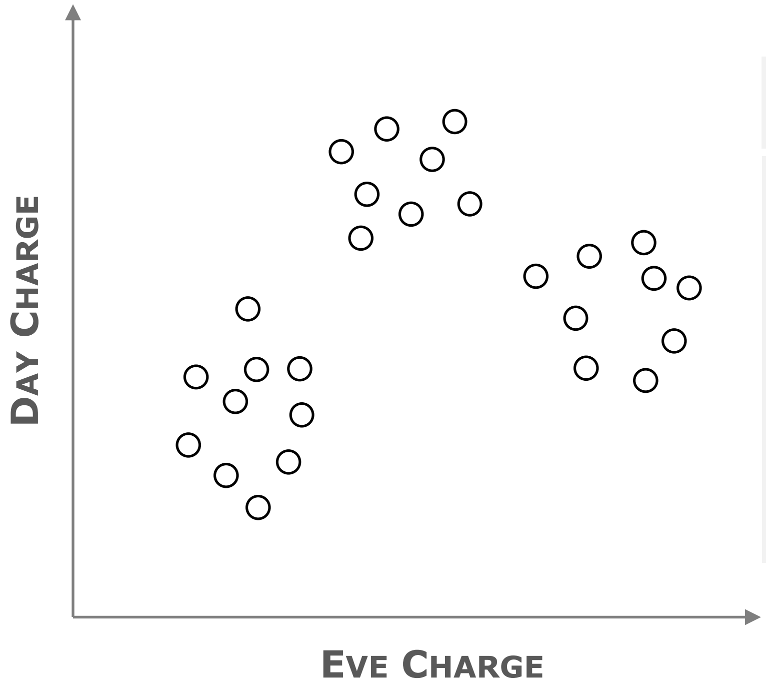

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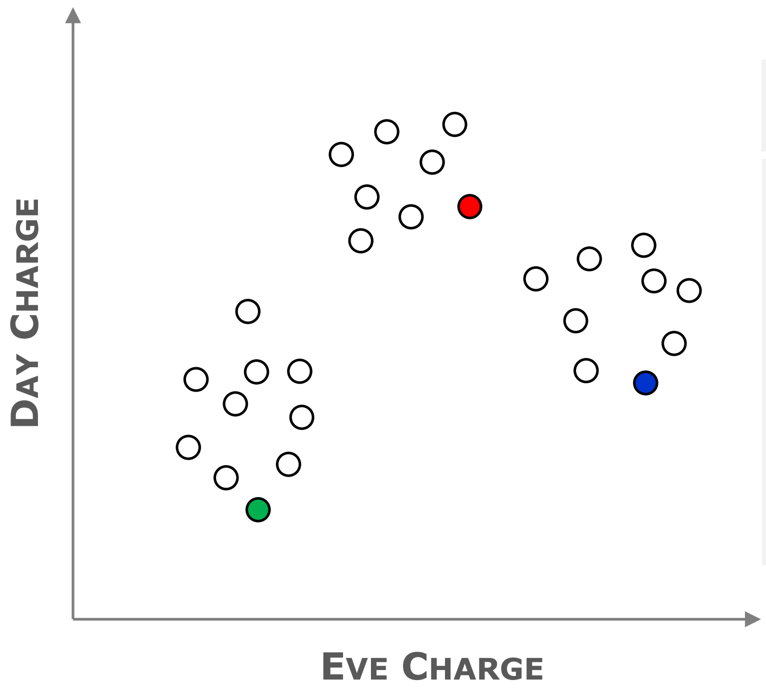

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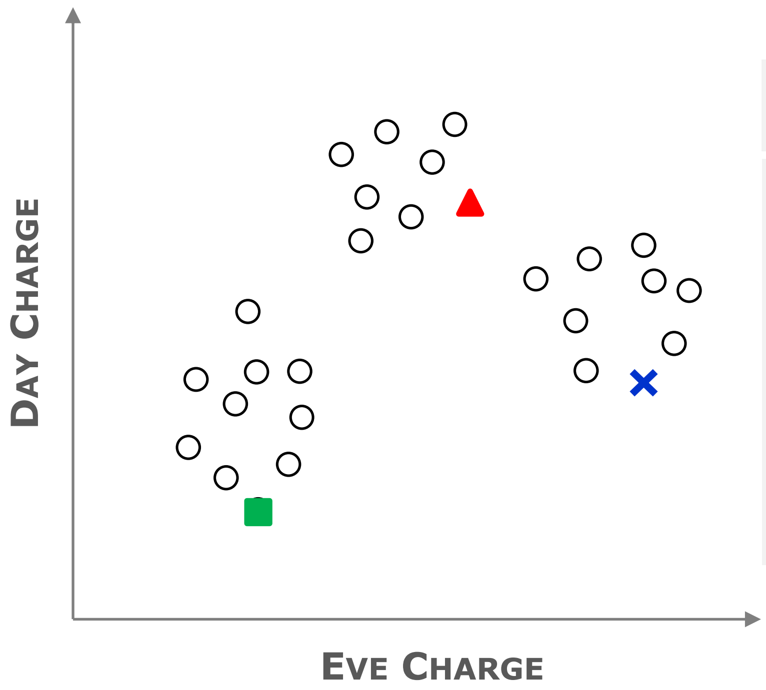

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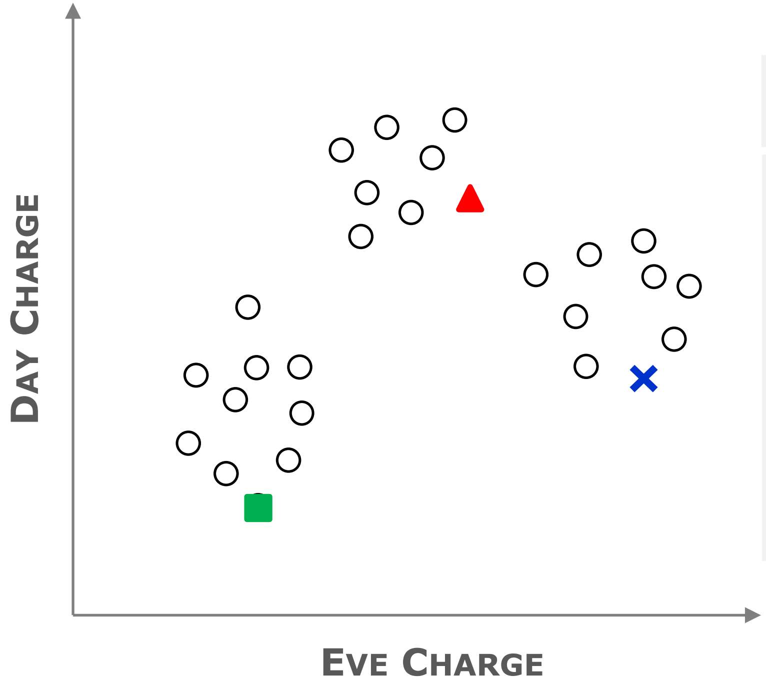

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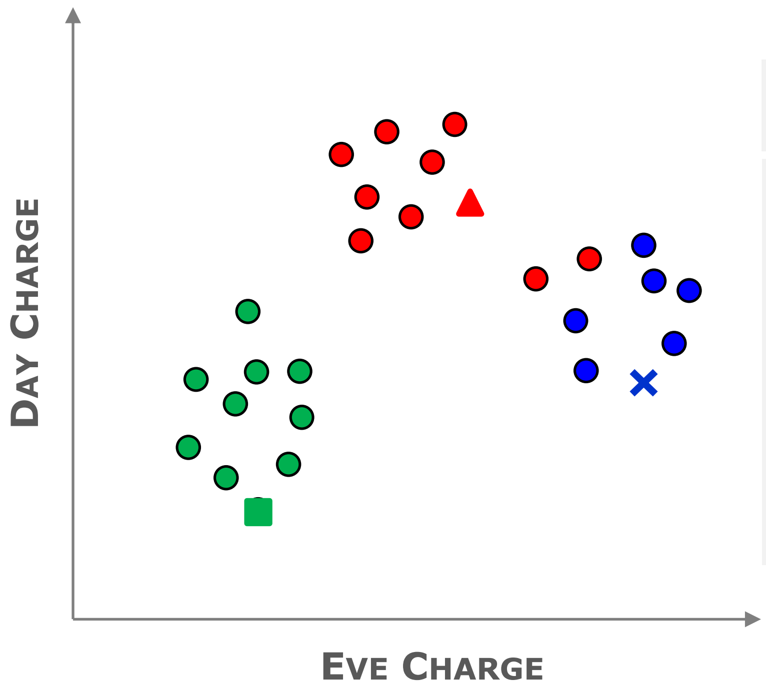

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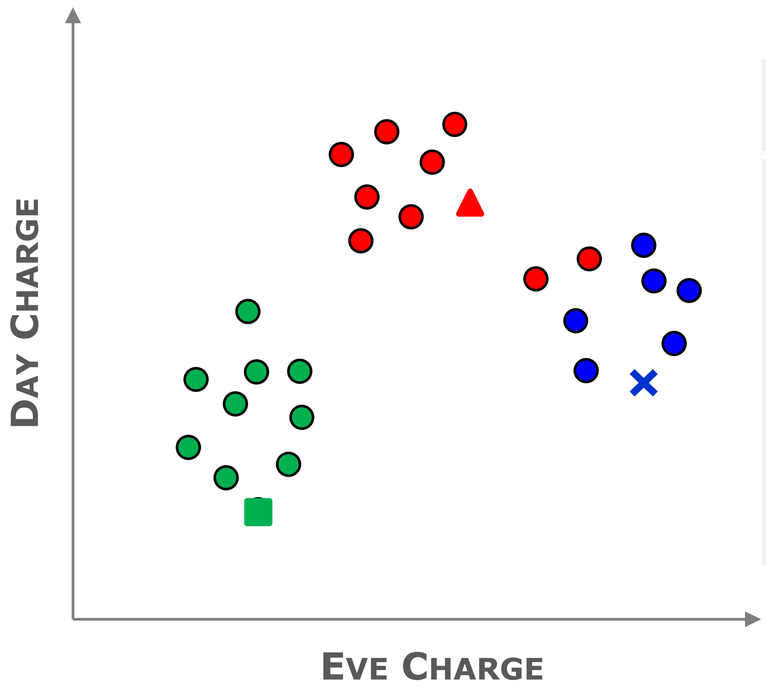

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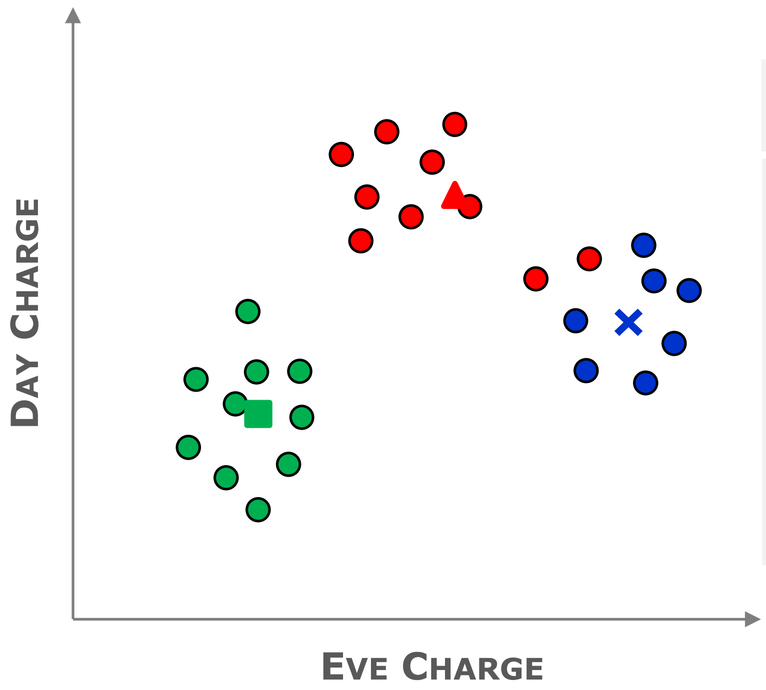

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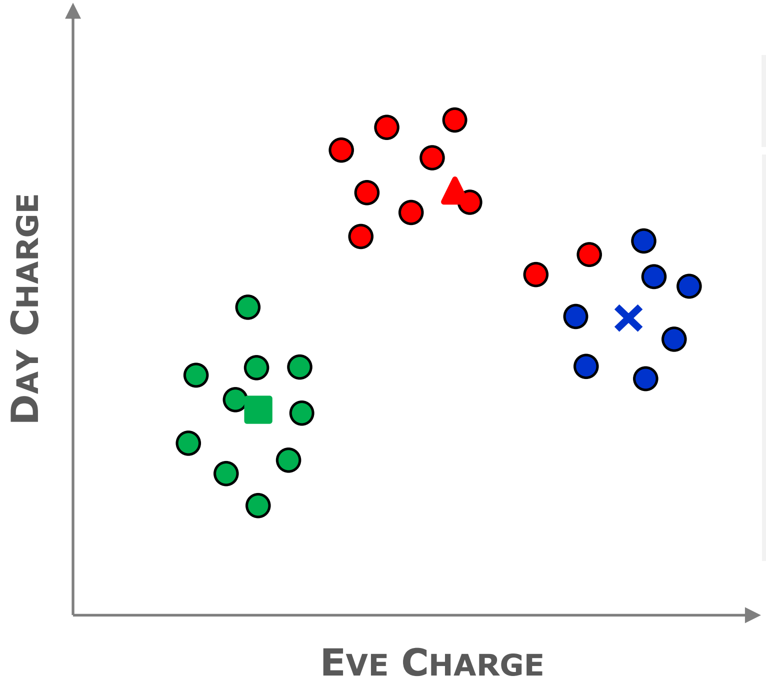

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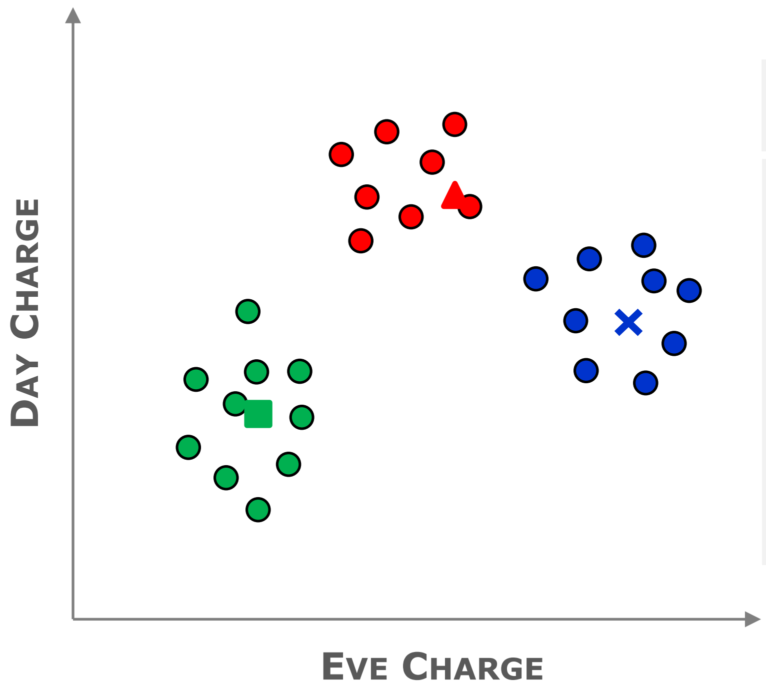

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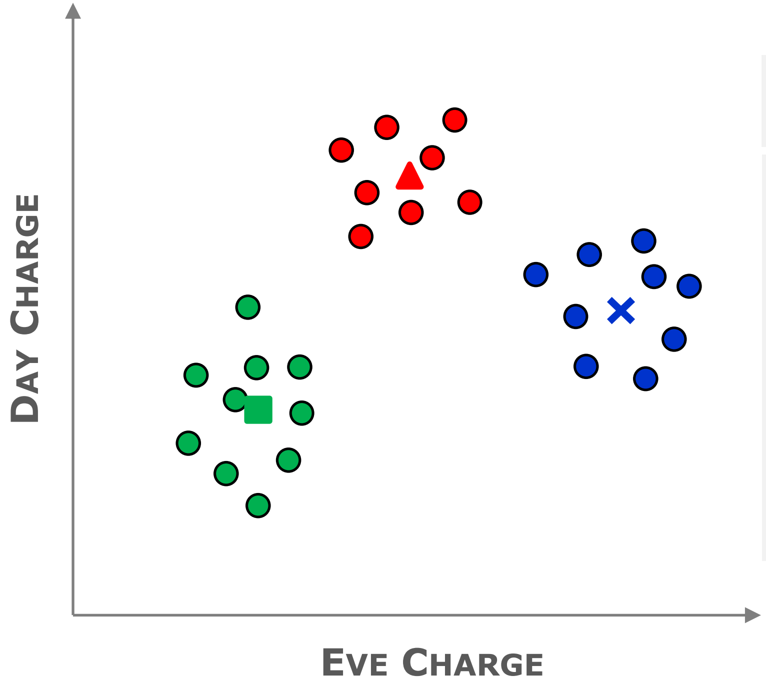

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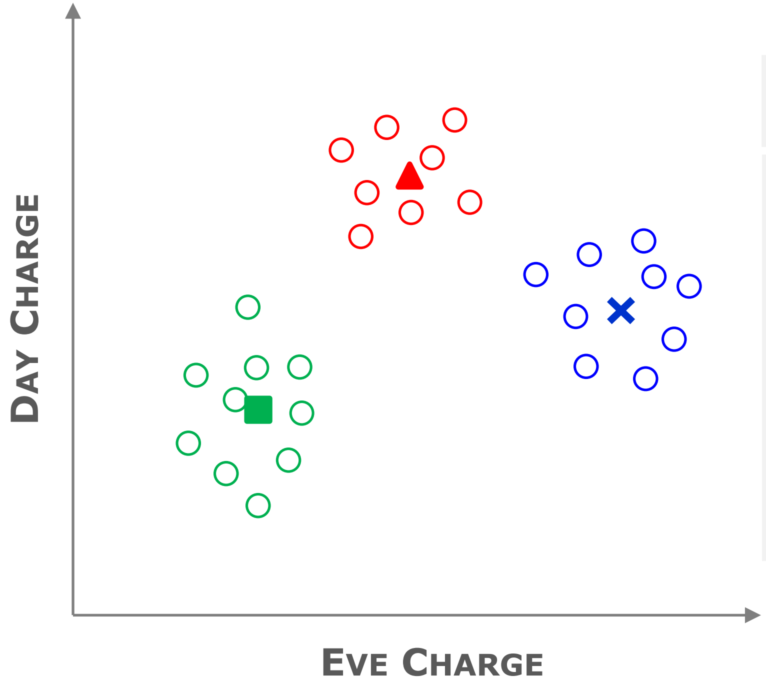

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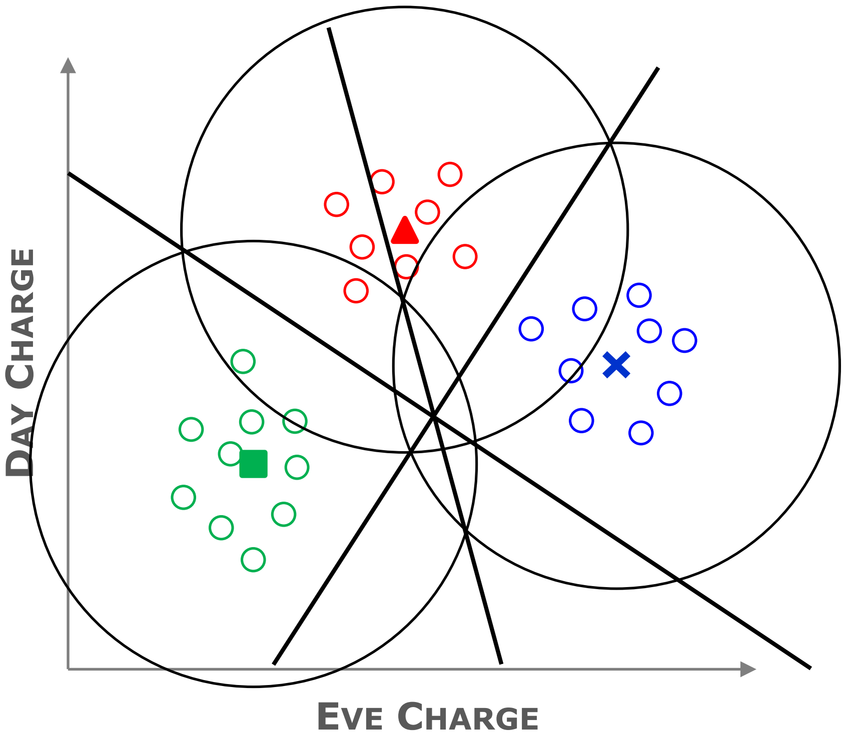

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